

Sequence-Derived Test Case Tables

Unit • Bottom-Up Integration • System End-to-End • Robustness

UC1–UC9 | Source of truth: corrected Mermaid sequence diagrams

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1. Testing Strategy and Conventions

These tables are derived strictly from sequence_diagrams_mermaid.txt. When an implementation or existing automated test differs, the sequence-diagram expectation remains the expected result and the discrepancy is marked as a conflict. No conflict was identified during this audit; missing and partial coverage are reported separately.

Testing direction: unit tests move from the UI boundary inward; bottom-up integration starts at models/services and then adds controllers/routes; system tests exercise the complete user-visible flow; robustness tests inject validation, dependency and state-integrity faults.

Coverage status legend

Status	Meaning
Implemented	A current automated test directly asserts the sequence-derived behavior.
Partially implemented	A related automated test exists, but omits part of the branch, boundary or real integration.
Not implemented	No matching automated test was found in the repository.
Conflicts with sequence diagram	A current test expects behavior that differs from the corrected sequence diagram.

General test-data rules

- Use synthetic student records and generated scan fixtures only.
- Never expose or assert raw scan filesystem paths in client responses.
- Preserve child-written text exactly. Dismiss errors; never delete embedded errors.
- Mock Gemini, Azure Storage and Resend for deterministic automated tests unless a test is explicitly labelled live external.
- Use an isolated MongoDB database for true persistence integration tests.

2. Unit Test Tables

Each table isolates one UI/controller/service/
model unit. Mock columns identify dependencies outside the target boundary. Persistent state is None when the unit does not own persistence.

UC1 — Upload Student Writing Sample for Analysis

Primary endpoint: POST /api/samples/:studentId

Current automated coverage — Implemented: 3; Not implemented: 5.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC1-01 UploadSamplePage: valid submit	studentId; title; taskType; valid files	Calls uploadSample and enters analysing phase	uploadSample(FormData)	202 Sample with status UPLOADED	None	None
UT-UC1-02 UploadSamplePage: rejected upload	Invalid file or unknown student	Shows inline error; retains entered form	uploadSample(FormData)	Reject with 400/422 error	None	None
UT-UC1-03 useSamplePolling: terminal outcomes	sampleId; repeated statuses	Completes on ANALYSED/REVIEWED; fails on FAILED; times out after 30 attempts	getSample(sampleId)	Status sequence	None	None
UT-UC1-04 createSample: accepted upload	req.files; studentId; taskType	Creates Sample; responds 202; starts runAnalysis after response	Student.findById; Sample.create; runAnalysis	student; created Sample; resolved job	Samples = []	Samples = [UPLOADED sample]
UT-UC1-05 createSample: invalid student/file	Unknown student or invalid second file	Returns 400/422; cleans temporary files; creates no Sample	Student.findById; toPageFiles; fs.unlink	null or validation error; unlink succeeds	Samples = []	Samples = []
UT-UC1-06 analyseSample: valid structured result	Sample pages and task context	Returns validated errors; preserves written text; uncertainty stays confidence-based	Gemini generateContent	Valid structured JSON	Sample = UPLOADED	Sample unchanged by pure analysis call
UT-UC1-07 Gemini retry and validation	Transient failure; malformed JSON/schema; timeout	Retries to configured limit; throws final controlled failure	Gemini generateContent	Failure then success, or repeated failure	Sample = UPLOADED	Sample unchanged by callModelWithRetry
UT-UC1-08 runAnalysis: persist outcome	sampleId	Saves ANALYSED with errors or FAILED with safe analysisError	Sample.findById; analyseSample; sample.save	Sample; success or failure; saved Sample	Sample.status = UPLOADED	Sample.status = ANALYSED or FAILED

UC2 — View Categorised Error Pattern Report

Primary endpoint: GET /api/samples/:sampleId

Current automated coverage — Implemented: 3; Not implemented: 4.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC2-01 SampleReportPage: group report	Serialized Sample with live and dismissed errors	Groups live errors in category order; omits empty categories	getSample; getStudent; getLatestRecommendations	Sample; student; null	None	None
UT-UC2-02 ErrorCard: uncertainty state	confidenceScore below threshold	Renders Uncertain and confirms through confidence update	onConfirm	callback captured	None	None
UT-UC2-03 ScanViewer: page overlays	Multiple pages; indexed boxes	Shows current-page boxes; switches pages; returns errorIndex	sampleImageUrl	Synthetic image URL	None	None
UT-UC2-04 getSample: found	sampleId	Returns toClientSample with statistics and no private imagePath	Sample.findById	Sample document	Sample exists	No state change
UT-UC2-05 getSample: missing	Unknown sampleId	Returns 404 Sample not found	Sample.findById	null	Sample absent	Sample absent
UT-UC2-06 getImages: valid and invalid index	sampleId; page index	Valid index sends bytes; invalid index returns 404; path never enters JSON	Sample.findById; res.sendFile	Sample with private path; sendFile result	Sample exists	No state change
UT-UC2-07 toClientSample: statistics	Sample with dismissed/live/unsure errors	Counts non-dismissed errors; keeps unsure in total; exposes confidenceThreshold	None	None	Sample exists	No state change

UC3 — View Recommended Intervention Strategies with Rationale

Primary endpoint: GET /api/students/:studentId/
recommendations/latest

Current automated coverage — Implemented: 1; Partially implemented: 2; Not implemented: 3.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC3-01 RecommendationsPage: existing report	studentId; current report	Displays strategies, evidence and rationale without generation	getLatestRecommendations; getStudent; getStudentTrends	report; student; trends	None	None
UT-UC3-02 RecommendationsPage: no report	studentId; latest returns null	Shows Generate recommendations or Upload sample state	getLatestRecommendations	null	None	None
UT-UC3-03 RecommendationsPage: outdated report	report.isOutdated = true	Shows Refresh now warning; does not auto-generate	getLatestRecommendations; generateRecommendations spy	outdated report; unused spy	None	None
UT-UC3-04 getLatest: current report	Valid studentId	Returns 200 report with isOutdated false; performs no write	RecommendationReport. findOne; Sample.exists	report; null	Report exists	Report unchanged
UT-UC3-05 getLatest: stale evidence	Valid studentId; changed/new Sample	Returns same report with computed isOutdated true	RecommendationReport. findOne; Sample.exists	report; matching Sample	Report exists	Report unchanged
UT-UC3-06 getLatest: invalid/missing report	Malformed ID or no report	Returns 400 INVALID_ID or 404 RECOMMENDATION_NOT_FOUND	RecommendationReport. findOne	not called or null	Report absent	Report absent

UC4 — Compare Error Trends Across Multiple Writing Samples

Primary endpoint: GET /api/students/:studentId/
trends

Current automated coverage — Implemented: 4; Partially implemented: 2; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC4-01 prepareTrendSamples	Unsorted server trends with unsure counts	Sorts chronologically; charts five categories; retains all-category total	None	None	None	None
UT-UC4-02 Range and sample filters	Prepared samples; preset/custom range; excluded IDs	Returns only included samples in range	None	None	None	None
UT-UC4-03 summariseTrends	Two or more included samples	Returns totals, most frequent category and improving/steady/ more-errors comparison	None	None	None	None
UT-UC4-04 ErrorTrendsPage: insufficient data	Zero or one included sample	Shows insufficient trend guidance	getStudentTrends; getStudent	0/1 trend samples; student	None	None
UT-UC4-05 getTrends: valid student	studentId; optional valid dates	Returns chronological per-sample counts excluding dismissed errors	Student.findById; Sample.find.sort	student; analysed/reviewed Samples	Samples exist	No state change
UT-UC4-06 getTrends: invalid dates	Malformed/impossible/ from-after-to dates	Returns 400 validation error; no Sample query	Student.findById; Sample.find spy	student; unused	No state change	No state change
UT-UC4-07 getTrends: missing student	Unknown studentId	Returns 404 Student not found	Student.findById	null	Student absent	Student absent

UC5 — Edit, Remove, Confirm, or Manually Add an Error

Primary endpoint: PATCH /api/samples/:sampleId/
errors/:errorIndex

Current automated coverage — Implemented: 7; Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC5-01 ErrorCard: reclassify	Existing error; one of five educator categories	Calls save with chosen category	onReclassify	captured callback	Error.category = old	Rendered card shows new category
UT-UC5-02 ErrorCard: remove/restore	Existing live or dismissed error	Confirms dismissal or offers restore	onDismiss/onRestore	captured callback	dismissed = false/true	Rendered state toggles
UT-UC5-03 ErrorCard: confirm uncertainty	Below-threshold confidence	Calls confirmation without changing category	onConfirm	captured callback	confidence < threshold	Rendered error is confirmed
UT-UC5-04 SampleReportPage: add missed error	written text; selected category	Calls updateSampleError(sample Id, new, payload)	updateSampleError	updated Sample	Form ready	Form submitted
UT-UC5-05 reclassifyError: category patch	sampleId; numeric index; valid category	Loads Sample first; changes category; saves once; returns full Sample	Sample.findById; sample.save	Sample; saved Sample	category = old	category = new
UT-UC5-06 reclassifyError: dismiss patch	sampleId; index; dismissed boolean	Preserves category; saves once; returns full Sample	Sample.findById; sample.save	Sample; saved Sample	dismissed = false	dismissed = true
UT-UC5-07 reclassifyError: confidence patch	sampleId; index; confidenceScore = 1	Preserves category/dismissal; saves confirmed score	Sample.findById; sample.save	Sample; saved Sample	confidence < threshold	confidenceScore = 1
UT-UC5-08 reclassifyError: append manual error	errorIndex = new; exact written; valid category	Appends active confidence=1 error; recomputes statistics	Sample.findById; sample.save	Sample; saved Sample	errors = prior list	errors += exact educator error
UT-UC5-09 reclassifyError: invalid request	Missing Sample; bad index/category/type/range/text	Returns 400/404 and does not save	Sample.findById; sample.save spy	null or Sample; unused save	Sample unchanged	Sample unchanged

UC6 — Regenerate Intervention Recommendations

Primary endpoint: POST /api/students/:studentId/recommendations

Current automated coverage — Implemented: 6; Not implemented: 3.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC6-01 Recommendation UI: generate/refresh	studentId; button click	Calls generateRecommendations once; replaces displayed report	generateRecommendations	201 current report	Old/no report displayed	New report displayed
UT-UC6-02 generateReport: active evidence	Student; analysed/reviewed Samples	Excludes dismissed errors; requests up to four strategies; returns sources	RecommendationEngine.createInterventionStrategies	Grounded strategies	No report mutation	No report mutation
UT-UC6-03 generateReport: no active errors	Samples absent or every error dismissed	Throws 422 NO_ACTIVE_ERRORS; engine not called	RecommendationEngine spy	unused	Existing report remains	Existing report remains
UT-UC6-04 RecommendationEngine: valid strategy	Evidence and approved worksheets	Returns validated grounded strategy list	Azure/Gemini adapters	Approved context; structured model output	None	None
UT-UC6-05 RecommendationEngine: external failure	Timeout/transient/malformed model response	Retries transient failure; returns controlled error for malformed output	Azure/Gemini adapters	Failure sequence	None	None
UT-UC6-06 generateStudentRecommendations: success	Valid studentId with active evidence	Generates first, then performs one atomic latest-only upsert; returns 201	Student.findById; Sample.find; generateReport; RecommendationReport.findOneAndUpdate	student; samples; generated report; current report	Old/no report	One current report
UT-UC6-07 generateStudentRecommendations: missing student	Unknown studentId	Returns 404 STUDENT_NOT_FOUND; no generation or upsert	Student.findById	null	No report change	No report change
UT-UC6-08 Concurrent first-upsert race	findOneAndUpdate throws E11000	Reads and returns winning current report	RecommendationReport.findOneAndUpdate; findOne	E11000; winner	Concurrent creation	One current report
UT-UC6-09 Generation failure preserves old report	Existing report; engine/generateReport rejects	Returns error before persistence; old report unchanged	generateReport; findOneAndUpdate spy	controlled rejection; unused upsert	Old report exists	Old report unchanged

UC7 — Log In

Primary endpoint: POST /api/auth/login

Current automated coverage — Implemented: 3; Partially implemented: 1; Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC7-01 LoginPage: successful submit	username; password	Saves safe account and navigates to requested route/caseload	login; saveSession	safe account; resolved	No client session	Account stored in localStorage
UT-UC7-02 LoginPage: rejected credentials	Unknown user or wrong password	Shows same inline error; does not navigate/save	login; saveSession spy	401 error; unused save	No client session	No client session
UT-UC7-03 authController.login: success	Known username; correct password	Returns safe account only; no JWT/reset/hash fields	Account.findOne; bcrypt.compare	account; true	Account exists	Account unchanged
UT-UC7-04 authController.login: missing input	Missing username or password	Returns 400 validation error	Account.findOne spy	unused	Account unchanged	Account unchanged
UT-UC7-05 authController.login: bad credentials	Unknown username or wrong password	Returns identical 401. Incorrect username or password	Account.findOne; bcrypt.compare	null/Account; false	Account unchanged	Account unchanged
UT-UC7-06 session route gate	safe account or no local session	saveSession persists object; RequireAuth permits or redirects	localStorage	stored account or null	Session absent/present	Session present/unchanged

UC8 — Change Password

Primary endpoint: PATCH /api/auth/change-password

Current automated coverage — Implemented: 1; Not implemented: 5.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC8-01 ChangePasswordForm: valid submit	current; strong new; matching confirmation	Calls changePassword; clears fields; shows Password updated	changePassword	200 success	Fields filled	Fields cleared
UT-UC8-02 ChangePasswordForm: invalid client input	Weak new password or mismatch	Disables submit or shows mismatch; no API call	changePassword spy	unused	Account unchanged	Account unchanged
UT-UC8-03 passwordPolicy: strength matrix	Passwords at/around each rule boundary	Returns all unmet rules; accepts exact valid boundary	None	None	None	None
UT-UC8-04 changePassword: valid	Known username; correct current; strong new	Hashes new password; saves once; returns success	Account.findOne; bcrypt.compare/hash; account.save	Account; true; new hash; saved	Old passwordHash	New passwordHash
UT-UC8-05 changePassword: wrong current	Unknown account or wrong current password	Returns same 401; does not hash/save	Account.findOne; bcrypt.compare; hash/save spies	null/Account; false; unused	Old passwordHash	Old passwordHash
UT-UC8-06 changePassword: weak new	Correct current; weak new password	Returns 400 requirements; does not hash/save	Account.findOne; bcrypt.compare; hash/save spies	Account; true; unused	Old passwordHash	Old passwordHash

UC9 — Reset a Forgotten Password

Primary endpoint: POST /api/auth/forgot-password + POST /api/auth/reset-password/
:token

Current automated coverage — Implemented: 5; Not implemented: 4.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
UT-UC9-01 ForgotPasswordPage: accepted	username	Shows Check your email	requestPasswordReset	200 generic response	phase = form	phase = sent
UT-UC9-02 ForgotPasswordPage: request error	username; provider/server rejection	Shows inline error; remains form	requestPasswordReset	request error	phase = form	phase = form
UT-UC9-03 forgotPassword: known account	Known username	Creates 64-hex token with 1-hour expiry; saves; sends link; returns generic 200	Account.findOne; account.save; sendPasswordResetEmail	Account; saved; email accepted	No reset token	Valid token + expiry
UT-UC9-04 forgotPassword: unknown account	Unknown username	Returns identical generic 200; sends no email	Account.findOne; send email spy	null; unused	No account	No account
UT-UC9-05 ResetPasswordPage: client validation	Weak or mismatched password	Disables submit or shows guidance; API not called	resetPassword spy	unused	phase = form	phase = form
UT-UC9-06 resetPassword: valid token	Unexpired token; strong password	Hashes password; clears token fields; saves; returns 200	Account.findOne; bcrypt.hash; account.save	Account; hash; saved	Token valid; old hash	Token null; expiry null; new hash
UT-UC9-07 resetPassword: weak password	Valid token; weak password	Returns 400 before token lookup/consumption	Account.findOne spy	unused	Token valid	Token remains valid
UT-UC9-08 resetPassword: invalid/expired token	Unknown or expired token; strong password	Returns 400 invalid/expired; no hash/save	Account.findOne; bcrypt/hash spies	null; unused	No valid token	No state change
UT-UC9-09 emailService: Resend adapter	Recipient and resetLink	Posts expected email; throws controlled provider failure	fetch(Resend API)	2xx JSON or non-2xx body	None	None

3. Bottom-Up Integration Test Tables

Each use case now uses five cumulative plans. Plan 1 proves the lowest meaningful backend collaboration. Plans 2–5 preserve every earlier real component and add exactly one logical layer: controller, HTTP transport, client API, then presentation. External AI/email boundaries remain mocked; MongoDB and synthetic files are real test dependencies.

UC1 — Upload Student Writing Sample for Analysis

Primary endpoint: POST /api/samples/:studentId

Integration Plan 1 — Analysis core

Real Components: errorClassificationEngine, Sample model, isolated MongoDB, synthetic page files

Mock Components: Gemini API only

Layer added in this plan: Analysis core

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC1-01 Persist successful analysis	UPLOADED sample; synthetic pages	Engine validates result and saves ANALYSED	Gemini.generateContent	Valid structured result	Sample = UPLOADED	Sample = ANALYSED with exact written errors
BI-UC1-02 Persist failed analysis	UPLOADED sample; repeated model failure	Bounded retry; safe reason; save FAILED	Gemini.generateContent	Failures through retry limit	Sample = UPLOADED	Sample = FAILED; no invalid errors

Integration Plan 2 — Upload controller + (Analysis core)

Real Components: errorClassificationEngine, Sample model, isolated MongoDB, synthetic page files, sampleController.
createSample

Mock Components: Gemini API only

Layer added in this plan: Upload controller

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC1-03 Controller starts proven core	Valid student; title/task; synthetic files	202 UPLOADED, then real core reaches ANALYSED	Gemini.generateContent	Valid structured result	Samples = []	One ANALYSED sample
BI-UC1-04 Controller preserves failed outcome	Valid upload; model failures	202 first; background core ends FAILED	Gemini.generateContent	Retry exhaustion	Samples = []	One FAILED sample with safe error

Integration Plan 3 — HTTP upload transport + (Upload controller + Analysis core)

Real Components: errorClassificationEngine, Sample model, isolated MongoDB, synthetic page files, sampleController.
createSample, samples route, Multer middleware

Mock Components: Gemini API only

Layer added in this plan: HTTP upload transport

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC1-05 Multipart route to proven core	Multipart valid fields/files	HTTP 202; later GET returns ANALYSED	Gemini.generateContent	Valid structured result	Samples = []	One ANALYSED sample
BI-UC1-06 Transported failure remains terminal	Multipart valid upload; model failures	HTTP 202; later GET returns FAILED	Gemini.generateContent	Retry exhaustion	Samples = []	One FAILED sample

Integration Plan 4 — Client API + (HTTP upload transport + Upload controller + Analysis core)

Real Components: errorClassificationEngine, Sample model, isolated MongoDB, synthetic page files, sampleController.
createSample, samples route, Multer middleware, uploadSample/getSample API functions

Mock Components: Gemini API only

Layer added in this plan: Client API

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC1-07 API client preserves upload contract	uploadSample(FormData), then getSample	Resolves 202 sample; polling sees ANALYSED	Gemini.generateContent	Valid structured result	Samples = []	One ANALYSED sample
BI-UC1-08 API client propagates validation	Invalid multipart data	Rejects with 400/422; no persisted sample	Gemini API	Not called	Samples = []	Samples = []

Integration Plan 5 — Presentation + (Client API + HTTP upload transport + Upload controller + Analysis core)

Real Components: errorClassificationEngine, Sample model, isolated MongoDB, synthetic page files, sampleController.
createSample, samples route, Multer middleware, uploadSample/getSample API functions, UploadSamplePage, useSamplePolling

Mock Components: Gemini API only

Layer added in this plan: Presentation

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC1-09 UI full integration success	Teacher submits valid synthetic page	Analysing state then success/report link	Gemini.generateContent	Valid structured result	Samples = []	One ANALYSED sample
BI-UC1-10 UI full integration failure	Teacher submits; analysis exhausts retry	Human-readable failure; form/profile recovery remains available	Gemini.generateContent	Retry exhaustion	Samples = []	One FAILED sample

UC2 — View Categorised Error Pattern Report

Primary endpoint: GET /api/samples/:sampleId

Integration Plan 1 — Sample read core

Real Components: Sample model, isolated MongoDB, toClientSample/
computeStatistics, synthetic scan files

Mock Components: None

Layer added in this plan: Sample read core

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC2-01 Read serialized live report	ANALYSED sample with live/dismissed errors	Serialized sample; live statistics; no imagePath	None	None	Sample exists	No mutation
BI-UC2-02 Read synthetic image safely	Sample page index; temp scan file	Image bytes obtained while path stays private	None	None	Sample exists with private path	No mutation

Integration Plan 2 — Read/image controller + (Sample read core)

Real Components: Sample model, isolated MongoDB, toClientSample/
computeStatistics, synthetic scan files, sampleController.
getSample/getImages

Mock Components: None

Layer added in this plan: Read/image controller

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC2-03 Controller returns report and image	Existing sample and valid page index	200 report plus safe image response	None	None	Sample exists	No mutation
BI-UC2-04 Controller handles missing resource	Unknown sample or bad page index	404 without path disclosure	None	None	Database unchanged	Database unchanged

Integration Plan 3 — HTTP read transport + (Read/image controller + Sample read core)

Real Components: Sample model, isolated MongoDB, toClientSample/
computeStatistics, synthetic scan files, sampleController.
getSample/getImages, sample and image routes

Mock Components: None

Layer added in this plan: HTTP read transport

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC2-05 GET routes preserve read core	GET report then image endpoint	HTTP 200 report and image bytes	None	None	Sample exists	No mutation
BI-UC2-06 GET routes preserve 404 branch	GET unknown sample/page	HTTP 404 safe error	None	None	Database unchanged	Database unchanged

Integration Plan 4 — Client API + (HTTP read transport + Read/image controller + Sample read core)

Real Components: Sample model, isolated MongoDB, toClientSample/
computeStatistics, synthetic scan files, sampleController.
getSample/getImages, sample and image routes, getSample/sampleImageUrl API functions

Mock Components: None

Layer added in this plan: Client API

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC2-07 API client reads integrated report	getSample; sampleImageUrl	Client receives report and fetchable image URL	None	None	Sample exists	No mutation
BI-UC2-08 API client propagates missing report	getSample(unknown id)	Rejected request with 404 meaning	None	None	Database unchanged	Database unchanged

Integration Plan 5 — Presentation + (Client API + HTTP read transport + Read/image controller + Sample read core)

Real Components: Sample model, isolated MongoDB, toClientSample/
computeStatistics, synthetic scan files, sampleController.
getSample/getImages, sample and image routes, getSample/sampleImageUrl API functions, SampleReportPage, ScanViewer, ErrorCard

Mock Components: None

Layer added in this plan: Presentation

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC2-09 UI renders integrated report	Teacher opens existing report	Groups live errors; uncertain state; scan boxes	None	None	Sample exists	No mutation
BI-UC2-10 UI renders missing state	Teacher opens unknown report	Shows Sample not found; no private path	None	None	Database unchanged	Database unchanged

UC3 — View Recommended Intervention Strategies with Rationale

Primary endpoint: GET /api/students/:studentId/
recommendations/latest

Integration Plan 1 — Recommendation data core

Real Components: RecommendationReport model, Sample model, isolated MongoDB

Mock Components: None

Layer added in this plan: Recommendation data core

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC3-01 Read latest report data	Student with latest report	Real query returns one latest-only report	None	None	Report exists	No mutation
BI-UC3-02 Detect changed evidence	Report generatedAt; newer usable sample	Real Sample.exists query finds freshness match	None	None	Report and newer sample exist	No mutation

Integration Plan 2 — Latest-report controller + (Recommendation data core)

Real Components: RecommendationReport model, Sample model, isolated MongoDB, getLatestStudent
Recommendations, freshness calculation

Mock Components: None

Layer added in this plan: Latest-report controller

Current automated coverage — Partially implemented: 2; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC3-03 Controller returns current report	Current report; no newer sample	200 report; isOutdated false; no write	None	None	Report exists	No mutation
BI-UC3-04 Controller returns no-report branch	Student has no recommendation	404 RECOMMENDATION_NOT_FOUND	None	None	Reports = []	Reports = []
BI-UC3-05 Controller computes outdated report	Report plus changed/new sample	200 report with isOutdated true; no write	None	None	Report exists	No mutation

Integration Plan 3 — HTTP latest-report transport + (Latest-report controller + Recommendation data core)

Real Components: RecommendationReport model, Sample model, isolated MongoDB, getLatestStudent
Recommendations, freshness calculation, recommendation latest route

Mock Components: None

Layer added in this plan: HTTP latest-report transport

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC3-06 GET latest current report	Valid latest-report HTTP request	HTTP 200 report; read-only	None	None	Report exists	No mutation
BI-UC3-07 GET latest no-report response	Valid student; no report	HTTP 404 expected code	None	None	Reports = []	Reports = []

Integration Plan 4 — Client API + (HTTP latest-report transport + Latest-report controller + Recommendation data core)

Real Components: RecommendationReport model, Sample model, isolated MongoDB, getLatestStudent
Recommendations, freshness calculation, recommendation latest route, getLatestRecommendations API function

Mock Components: None

Layer added in this plan: Client API

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC3-08 API maps current report	getLatestRecommendations(student Id)	Resolves report envelope	None	None	Report exists	No mutation
BI-UC3-09 API maps expected absence	getLatestRecommendations(student Id)	Maps RECOMMENDATION_NOT_FOUND to null	None	None	Reports = []	Reports = []

Integration Plan 5 — Presentation + (Client API + HTTP latest-report transport + Latest-report controller + Recommendation data core)

Real Components: RecommendationReport model, Sample model, isolated MongoDB, getLatestStudent
Recommendations, freshness calculation, recommendation latest route, getLatestRecommendations API function, RecommendationsPage

Mock Components: None

Layer added in this plan: Presentation

Current automated coverage — Not implemented: 3.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC3-10 UI displays current report	Teacher opens recommendations	Strategies, evidence and rationale shown	None	None	Report exists	No mutation
BI-UC3-11 UI displays no-report action	Teacher opens; no report	Generate/upload state shown	None	None	Reports = []	Reports = []
BI-UC3-12 UI displays outdated warning	Teacher opens stale report	Report retained; Refresh now warning shown	None	None	Report exists; newer sample exists	No mutation

UC4 — Compare Error Trends Across Multiple Writing Samples

Primary endpoint: GET /api/students/:studentId/trends

Integration Plan 1 — Trend data core

Real Components: Student model, Sample model, isolated MongoDB

Mock Components: None

Layer added in this plan: Trend data core

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC4-01 Query chronological usable samples	Existing student with mixed statuses	Only ANALYSED/REVIEWED samples sorted oldest first	None	None	Student and samples exist	No mutation
BI-UC4-02 No student data branch	Unknown student id	Real lookup returns no Student	None	None	Student absent	No mutation

Integration Plan 2 — Trend controller + (Trend data core)

Real Components: Student model, Sample model, isolated MongoDB, studentController.
getTrends, server aggregation

Mock Components: None

Layer added in this plan: Trend controller

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC4-03 Controller aggregates live errors	Valid student; chronological samples	Totals/per-sample trends exclude dismissed; keep unsure in total	None	None	Student/samples exist	No mutation
BI-UC4-04 Controller rejects invalid range	from/to malformed or reversed	400 date error; no sample query mutation	None	None	Database unchanged	Database unchanged

Integration Plan 3 — HTTP trends transport + (Trend controller + Trend data core)

Real Components: Student model, Sample model, isolated MongoDB, studentController.
getTrends, server aggregation, student trends route

Mock Components: None

Layer added in this plan: HTTP trends transport

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC4-05 GET trends integrated success	HTTP GET all-time or valid range	200 chronological trends payload	None	None	Student/samples exist	No mutation
BI-UC4-06 GET trends integrated errors	Unknown student or invalid range	HTTP 404/400 safe response	None	None	Database unchanged	Database unchanged

Integration Plan 4 — Client API + (HTTP trends transport + Trend controller + Trend data core)

Real Components: Student model, Sample model, isolated MongoDB, studentController.
getTrends, server aggregation, student trends route, getStudentTrends API function

Mock Components: None

Layer added in this plan: Client API

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC4-07 API requests all-time trends	getStudentTrends(student Id)	Resolves totals and trends once	None	None	Student/samples exist	No mutation
BI-UC4-08 API propagates not-found	getStudentTrends(unknown id)	Rejects with 404 meaning	None	None	Database unchanged	Database unchanged

Integration Plan 5 — Presentation + (Client API + HTTP trends transport + Trend controller + Trend data core)

Real Components: Student model, Sample model, isolated MongoDB, studentController.
getTrends, server aggregation, student trends route, getStudentTrends API function, ErrorTrendsPage, client trends helpers, TrendChart

Mock Components: None

Layer added in this plan: Presentation

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC4-09 UI renders comparison	Teacher opens; selects at least two samples	Five category lines, total summary and comparison	None	None	At least two usable samples	No mutation
BI-UC4-10 UI shows insufficient guidance	Filters leave fewer than two samples	No chart; guidance shown	None	None	Zero/one included sample	No mutation

UC5 — Edit, Remove, Confirm, or Manually Add an Error

Primary endpoint: PATCH /api/samples/:sampleId/
errors/:errorIndex

Integration Plan 1 — Embedded-error data core

Real Components: Sample model, isolated MongoDB, toClientSample/
computeStatistics

Mock Components: None

Layer added in this plan: Embedded-error data core

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC5-01 Persist existing-error mutation	Sample; valid error index and patch	Embedded error changes; statistics recomputed	None	None	Sample has baseline error	Sample retains error with updated fields
BI-UC5-02 Persist educator-added error	Sample; written text and category	Append exact written text; confidence 1; dismissed false	None	None	Sample has baseline errors	Sample includes new embedded error

Integration Plan 2 — Error-update controller + (Embedded-error data core)

Real Components: Sample model, isolated MongoDB, toClientSample/
computeStatistics, sampleController.
reclassifyError

Mock Components: None

Layer added in this plan: Error-update controller

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC5-03 Controller handles supported actions	Category/dismissed/ confidence/new patches	200 full sample; fresh live counts	None	None	Sample exists	Requested mutation saved
BI-UC5-04 Controller rejects invalid patch	Bad index/category/type/empty written	400/404; no mutation	None	None	Sample exists	Sample unchanged

Integration Plan 3 — HTTP PATCH transport + (Error-update controller + Embedded-error data core)

Real Components: Sample model, isolated MongoDB, toClientSample/
computeStatistics, sampleController.
reclassifyError, sample error route

Mock Components: None

Layer added in this plan: HTTP PATCH transport

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC5-05 PATCH route preserves mutations	HTTP PATCH for existing/new error	HTTP 200 full updated sample	None	None	Sample exists	Requested mutation saved
BI-UC5-06 PATCH route preserves validation	HTTP PATCH invalid payload/index	HTTP 400/404; unchanged sample	None	None	Sample exists	Sample unchanged

Integration Plan 4 — Client API + (HTTP PATCH transport + Error-update controller + Embedded-error data core)

Real Components: Sample model, isolated MongoDB, toClientSample/
computeStatistics, sampleController.
reclassifyError, sample error route, updateSampleError API function

Mock Components: None

Layer added in this plan: Client API

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC5-07 API sends all patch forms	updateSampleError existing/new	Resolves full updated sample	None	None	Sample exists	Requested mutation saved
BI-UC5-08 API propagates save rejection	Invalid patch or missing sample	Rejects with server error; no stale success	None	None	Sample unchanged	Sample unchanged

Integration Plan 5 — Presentation + (Client API + HTTP PATCH transport + Error-update controller + Embedded-error data core)

Real Components: Sample model, isolated MongoDB, toClientSample/
computeStatistics, sampleController.
reclassifyError, sample error route, updateSampleError API function, SampleReportPage, ErrorCard

Mock Components: None

Layer added in this plan: Presentation

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC5-09 UI integrates correction actions	Teacher reclassifies/removes/ restores/confirmes/adds	Groups/counts/card repaint from server response	None	None	Sample has baseline errors	Sample contains persisted educator decision
BI-UC5-10 UI preserves failure state	Teacher submits invalid/rejected change	Inline failure; displayed report not falsely changed	None	None	Sample and UI baseline agree	Persistent sample unchanged

UC6 — Regenerate Intervention Recommendations

Primary endpoint: POST /api/students/:studentId/recommendations

Integration Plan 1 — Recommendation generation core

Real Components: generateReport, RecommendationEngine, Student/Sample/RecommendationReport models, isolated MongoDB

Mock Components: Azure Storage and Gemini APIs only

Layer added in this plan: Recommendation generation core

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC6-01 Generate grounded report core	Student with active non-dismissed errors	Up to four grounded strategies; latest report data ready	Azure/Gemini APIs	Approved catalogue/context and valid strategies	Old report optional	Old report unchanged until controller persists
BI-UC6-02 Reject no active errors	Samples contain no active errors	422 NO_ACTIVE_ERRORS; no generated report	Azure/Gemini APIs	Not called	Old report optional	Old report unchanged

Integration Plan 2 — Generation controller + (Recommendation generation core)

Real Components: generateReport, RecommendationEngine, Student/Sample/RecommendationReport models, isolated MongoDB, generateStudentRecommendations, atomic latest-only upsert

Mock Components: Azure Storage and Gemini APIs only

Layer added in this plan: Generation controller

Current automated coverage — Partially implemented: 2; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC6-03 Controller atomically replaces latest	Valid student and generated report	One findOneAndUpdate upsert; 201 current report	Azure/Gemini APIs	Valid grounded strategies	Old report optional	Exactly one current report
BI-UC6-04 Controller retains report on failure	Generation rejects before persistence	Error response; no upsert	Azure/Gemini APIs	Generation failure	Old report exists	Old report unchanged
BI-UC6-05 Controller recovers duplicate race	Concurrent first upsert returns E11000	Reads winning current report; no history duplicate	Azure/Gemini APIs	Valid grounded strategies	No report initially	Exactly one winning report

Integration Plan 3 — HTTP generation transport + (Generation controller + Recommendation generation core)

Real Components: generateReport, RecommendationEngine, Student/Sample/RecommendationReport models, isolated MongoDB, generateStudentRecommendations, atomic latest-only upsert, recommendation POST route

Mock Components: Azure Storage and Gemini APIs only

Layer added in this plan: HTTP generation transport

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC6-06 POST route returns generated report	Valid student request	HTTP 201 current report	Azure/Gemini APIs	Valid grounded strategies	Old report optional	Exactly one current report
BI-UC6-07 POST route preserves failure	No errors or external failure	HTTP 422/error; old report retained	Azure/Gemini APIs	No evidence or failure	Old report exists	Old report unchanged

Integration Plan 4 — Client API + (HTTP generation transport + Generation controller + Recommendation generation core)

Real Components: generateReport, RecommendationEngine, Student/Sample/RecommendationReport models, isolated MongoDB, generateStudentRecommendations, atomic latest-only upsert, recommendation POST route, generateRecommendations API function

Mock Components: Azure Storage and Gemini APIs only

Layer added in this plan: Client API

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC6-08 API resolves generated report	generateRecommendations(studentId)	Resolves 201 report envelope	Azure/Gemini APIs	Valid grounded strategies	Old report optional	Exactly one current report
BI-UC6-09 API rejects without replacing client data	Generation request fails	Rejected promise; server report unchanged	Azure/Gemini APIs	Failure	Old report exists	Old report unchanged

Integration Plan 5 — Presentation + (Client API + HTTP generation transport + Generation controller + Recommendation generation core)

Real Components: generateReport, RecommendationEngine, Student/Sample/RecommendationReport models, isolated MongoDB, generateStudentRecommendations, atomic latest-only upsert, recommendation POST route, generateRecommendations API function, RecommendationsPage, SampleReportPage refresh trigger

Mock Components: Azure Storage and Gemini APIs only

Layer added in this plan: Presentation

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC6-10 UI replaces report after success	Teacher selects Generate/Refresh	Displayed report changes only after 201	Azure/Gemini APIs	Valid grounded strategies	Old report optional	Exactly one current report
BI-UC6-11 UI retains report after failure	Teacher refreshes; generation fails	Existing report remains; error shown	Azure/Gemini APIs	Failure	Old report exists	Old report unchanged

UC7 — Log In

Primary endpoint: POST /api/auth/login

Integration Plan 1 — Credential data core

Real Components: Account model, bcryptjs, isolated MongoDB

Mock Components: None

Layer added in this plan: Credential data core

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC7-01 Verify valid credentials core	Stored account; correct password	Real query and bcrypt comparison succeed	None	None	Account exists	No mutation
BI-UC7-02 Reject invalid credentials core	Missing account or wrong password	Same invalid result; no mutation	None	None	Account optional	No mutation

Integration Plan 2 — Login controller + (Credential data core)

Real Components: Account model, bcryptjs, isolated MongoDB, authController.login, safe-account serializer

Mock Components: None

Layer added in this plan: Login controller

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC7-03 Controller returns safe account	Valid username/password	200 object excludes hash/reset fields/JWT	None	None	Account exists	No mutation
BI-UC7-04 Controller returns generic rejection	Unknown user or wrong password	401 same message	None	None	Account optional	No mutation

Integration Plan 3 — HTTP login transport + (Login controller + Credential data core)

Real Components: Account model, bcryptjs, isolated MongoDB, authController.login, safe-account serializer, auth login route

Mock Components: None

Layer added in this plan: HTTP login transport

Current automated coverage — Partially implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC7-05 POST login success	HTTP credentials payload	HTTP 200 safe account	None	None	Account exists	No mutation
BI-UC7-06 POST login validation/rejection	Missing or invalid credentials	HTTP 400/401 safe error	None	None	Account optional	No mutation

Integration Plan 4 — Client API + (HTTP login transport + Login controller + Credential data core)

Real Components: Account model, bcryptjs, isolated MongoDB, authController.login, safe-account serializer, auth login route, login API function

Mock Components: None

Layer added in this plan: Client API

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC7-07 API resolves account	login({ username, password })	Resolves safe account only	None	None	Account exists	No mutation
BI-UC7-08 API propagates generic error	login(invalid credentials)	Rejects with same message	None	None	Account optional	No mutation

Integration Plan 5 — Presentation/session + (Client API + HTTP login transport + Login controller + Credential data core)

Real Components: Account model, bcryptjs, isolated MongoDB, authController.login, safe-account serializer, auth login route, login API function, LoginPage, session localStorage helper

Mock Components: None

Layer added in this plan: Presentation/session

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC7-09 UI saves session and navigates	Teacher submits valid credentials	Session stored; requested/default route opened	None	None	No client session	localStorage has safe account
BI-UC7-10 UI shows inline rejection	Teacher submits invalid credentials	Same inline error; no session	None	None	No client session	No client session

UC8 — Change Password

Primary endpoint: PATCH /api/auth/change-password

Integration Plan 1 — Password data core

Real Components: Account model, bcryptjs, passwordPolicy, isolated MongoDB

Mock Components: None

Layer added in this plan: Password data core

Current automated coverage — Partially implemented: 1; Not implemented: 1.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC8-01 Persist strong new password core	Account; correct current; compliant new password	Real bcrypt verify/hash and save succeeds	None	None	Account has old hash	Account has new hash
BI-UC8-02 Reject weak new password core	Account; weak new password	Policy problems; old hash retained	None	None	Account has old hash	Account has old hash

Integration Plan 2 — Change-password controller + (Password data core)

Real Components: Account model, bcryptjs, passwordPolicy, isolated MongoDB, authController.
changePassword

Mock Components: None

Layer added in this plan: Change-password controller

Current automated coverage — Partially implemented: 1; Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC8-03 Controller changes valid password	Valid username/current/new values	200 updated; saved hash matches new password	None	None	Account has old hash	Account has new hash
BI-UC8-04 Controller hides account/current mismatch	Missing account or wrong current password	401 same message; no save	None	None	Account optional	No hash change
BI-UC8-05 Controller enforces server policy	Valid current; weak new password	400 requirements; no save	None	None	Account has old hash	Account has old hash

Integration Plan 3 — HTTP password transport + (Change-password controller + Password data core)

Real Components: Account model, bcryptjs, passwordPolicy, isolated MongoDB, authController.
changePassword, auth change-password route

Mock Components: None

Layer added in this plan: HTTP password transport

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC8-06 PATCH password success	HTTP valid change payload	HTTP 200; persisted new hash	None	None	Account has old hash	Account has new hash
BI-UC8-07 PATCH password rejection	Wrong current or weak new value	HTTP 401/400; old hash retained	None	None	Account has old hash	Account has old hash

Integration Plan 4 — Client API + (HTTP password transport + Change-password controller + Password data core)

Real Components: Account model, bcryptjs, passwordPolicy, isolated MongoDB, authController.
changePassword, auth change-password route, changePassword API function

Mock Components: None

Layer added in this plan: Client API

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC8-08 API resolves change success	changePassword(valid values)	Resolves updated message	None	None	Account has old hash	Account has new hash
BI-UC8-09 API propagates safe error	changePassword(invalid values)	Rejects with 401/400 meaning	None	None	Account has old hash	Account has old hash

Integration Plan 5 — Presentation + (Client API + HTTP password transport + Change-password controller + Password data core)

Real Components: Account model, bcryptjs, passwordPolicy, isolated MongoDB, authController.
changePassword, auth change-password route, changePassword API function, AccountPage.
ChangePasswordForm, client password rules

Mock Components: None

Layer added in this plan: Presentation

Current automated coverage — Not implemented: 2.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC8-10 UI completes password change	Teacher enters matching valid values	Success shown; fields clear	None	None	Account has old hash	Account has new hash
BI-UC8-11 UI blocks or reports invalid values	Mismatch, weak new, or wrong current	Disabled/guidance or inline server error; fields retained	None	None	Account has old hash	Account has old hash

UC9 — Reset a Forgotten Password

Primary endpoint: POST /api/auth/forgot-password; POST /api/auth/reset-password/
:token

Integration Plan 1 — Reset data/service core

Real Components: Account model, passwordPolicy, bcryptjs, emailService, isolated MongoDB

Mock Components: Resend API only

Layer added in this plan: Reset data/service core

Current automated coverage — Partially implemented: 3.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC9-01 Persist reset token for known account	Known account; generated token/expiry	Token and future expiry saved	Resend API	Accepted email response	Account has no reset token	Account has token and expiry
BI-UC9-02 Keep unknown-account privacy core	Unknown username	No account/email mutation	Resend API	Not called	No matching account	No mutation
BI-UC9-03 Consume valid token core	Valid unexpired token; strong password	New hash saved; token/expiry cleared	Resend API	Not called	Account has valid token	Account has new hash; token null

Integration Plan 2 — Reset controllers + (Reset data/service core)

Real Components: Account model, passwordPolicy, bcryptjs, emailService, isolated MongoDB, authController.
forgotPassword/
resetPassword

Mock Components: Resend API only

Layer added in this plan: Reset controllers

Current automated coverage — Partially implemented: 3.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC9-04 Forgot controller sends generic success	Known or unknown username	Same 200 message; email only for known account	Resend API	Accepted for known account	Token absent	Known account token saved; unknown unchanged
BI-UC9-05 Reset controller updates valid token	Valid token; compliant password	200 updated; token cleared	Resend API	Not called	Account has valid token	Account has new hash; token null
BI-UC9-06 Reset controller rejects invalid/weak	Unknown/expired token or weak password	400; token not wrongly consumed	Resend API	Not called	Account/token optional	Persistent state unchanged

Integration Plan 3 — HTTP reset transport + (Reset controllers + Reset data/service core)

Real Components: Account model, passwordPolicy, bcryptjs, emailService, isolated MongoDB, authController.
forgotPassword/
resetPassword, auth forgot/reset routes

Mock Components: Resend API only

Layer added in this plan: HTTP reset transport

Current automated coverage — Partially implemented: 3.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC9-07 Forgot route privacy pair	POST known and unknown usernames	Both HTTP 200 generic; one email attempt	Resend API	Accepted for known account	Known account token absent	Known account token saved
BI-UC9-08 Reset route success	POST valid token and strong password	HTTP 200; token cleared	Resend API	Not called	Account has valid token	Account has new hash; token null
BI-UC9-09 Reset route rejection	POST weak password or invalid/expired token	HTTP 400; state retained	Resend API	Not called	Account/token optional	Persistent state unchanged

Integration Plan 4 — Client API + (HTTP reset transport + Reset controllers + Reset data/service core)

Real Components: Account model, passwordPolicy, bcryptjs, emailService, isolated MongoDB, authController.
forgotPassword/
resetPassword, auth forgot/reset routes, requestPasswordReset/
resetPassword API functions

Mock Components: Resend API only

Layer added in this plan: Client API

Current automated coverage — Not implemented: 3.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC9-10 API requests reset privately	requestPasswordReset(username)	Resolves same generic message	Resend API	Accepted for known account	Token absent	Known account token saved if found
BI-UC9-11 API submits valid reset	resetPassword(token,password)	Resolves updated message	Resend API	Not called	Account has valid token	Account has new hash; token null
BI-UC9-12 API propagates reset rejection	Weak password or invalid token	Rejects with 400 meaning	Resend API	Not called	Account/token optional	Persistent state unchanged

Integration Plan 5 — Presentation + (Client API + HTTP reset transport + Reset controllers + Reset data/service core)

Real Components: Account model, passwordPolicy, bcryptjs, emailService, isolated MongoDB, authController.
forgotPassword/
resetPassword, auth forgot/reset routes, requestPasswordReset/
resetPassword API functions, ForgotPasswordPage, ResetPasswordPage

Mock Components: Resend API only

Layer added in this plan: Presentation

Current automated coverage — Not implemented: 3.

Scenario / Target Method	Inputs	Expected Output	Mock Input	Mock Output	System State Before	System State After
BI-UC9-13 Forgot UI keeps privacy	Teacher requests known/unknown username	Same Check your email state	Resend API	Accepted for known account	Token absent	Known account token saved if found
BI-UC9-14 Reset UI completes valid flow	Teacher follows valid link; enters strong matching password	Success and sign-in link shown	Resend API	Not called	Account has valid token	Account has new hash; token null
BI-UC9-15 Reset UI handles validation/error	Mismatch, weak password, invalid/expired token	Guidance/inline error; no false success	Resend API	Not called	Account/token optional	Persistent state unchanged

4. System End-to-End Test Tables

System tests use a real browser, Vite client, Express server and isolated MongoDB. External services use deterministic test doubles. No automated browser E2E suite was found, so every case below is currently Not implemented.

UC1 — Upload Student Writing Sample for Analysis

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC1-01 Upload and analyse successfully	Signed-in teacher; student exists; synthetic valid page; deterministic AI	Open student → Upload sample → choose file/task → Analyse → wait	202 accepted; analysing screen; success state; Open error report enabled	No Sample	Sample = ANALYSED with errors	No automated browser E2E test found.
E2E-UC1-02 Reject invalid upload	Signed-in teacher; student exists; invalid synthetic file	Submit invalid upload	Inline 400/422 error; entered form retained; no Sample	No Sample	No Sample	No automated browser E2E test found.
E2E-UC1-03 Analysis failure and polling timeout states	Synthetic analysis failure or non-terminal polling fixture	Upload; observe failure. Repeat with stalled polling; choose Check again	FAILED shows safe reason; timeout offers Check again/return	No Sample	FAILED or still-UPLOADED test Sample	No automated browser E2E test found.

UC2 — View Categorised Error Pattern Report

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC2-01 View categorised report	ANALYSED synthetic Sample with live/dismissed/uncertain/multi-page errors	Open report; filter category; switch page; select box; reveal removed tags	Correct groups/counts/uncertainty/scan boxes; no raw imagePath	Sample exists	No persistent change	No automated browser E2E test found.
E2E-UC2-02 Missing report or image	Unknown sample or invalid synthetic image index	Open report/image URL	404 state or image fallback; no private path disclosed	Sample absent/invalid page	No state change	No automated browser E2E test found.

UC3 — View Recommended Intervention Strategies with Rationale

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC3-01 View current recommendations	Current RecommendationReport exists	Open Recommendations	Existing strategies/evidence/ rationale displayed; no POST generation	Current report exists	Same report unchanged	No automated browser E2E test found.
E2E-UC3-02 View no-report state	Analysed sample exists but report absent	Open Recommendations	Generate recommendations action displayed; no automatic POST	No report	No report	No automated browser E2E test found.
E2E-UC3-03 View outdated report	Report exists; source Sample updated later	Open Recommendations	Old report displayed with Refresh now warning; no automatic regeneration	Old report + changed Sample	Same report unchanged	No automated browser E2E test found.

UC4 — Compare Error Trends Across Multiple Writing Samples

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC4-01 Compare multiple samples	At least three analysed/reviewed samples with synthetic errors	Open trends; choose range; exclude/reinclude sample	Chart and summary update locally; totals reconcile with report counts	Samples exist	No persistent change	No automated browser E2E test found.
E2E-UC4-02 Insufficient comparison	Zero/one sample remains after filtering	Open trends; narrow range or exclude samples	Insufficient trend guidance; no misleading comparison	Samples exist	No persistent change	No automated browser E2E test found.
E2E-UC4-03 Missing student	Unknown student route	Open trends URL	Student not found shown	Student absent	Student absent	No automated browser E2E test found.

UC5 — Edit, Remove, Confirm, or Manually Add an Error

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC5-01 Correct existing AI tags	Report has reclassifiable, dismissible and uncertain errors	Reclassify one; remove one; confirm one	Each card/count updates from full server Sample; written text unchanged	Original embedded errors	Category/dismissed/confidence updated	No automated browser E2E test found.
E2E-UC5-02 Restore removed tag	Report has dismissed error	Show removed tags → Restore	Error returns to live group and counts	dismissed = true	dismissed = false	No automated browser E2E test found.
E2E-UC5-03 Add an AI-missed error	Report exists	Enter exact child-written text; select one of five categories; Add missed error	New educator error appears with confidence 1; counts increase	No manual error	Exact manual error appended	No automated browser E2E test found.
E2E-UC5-04 Rejected correction	Inject missing Sample or invalid patch response	Attempt correction	Inline failure; UI does not claim success	Sample/UI baseline	Baseline retained	No automated browser E2E test found.

UC6 — Regenerate Intervention Recommendations

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC6-01 Generate first recommendations	Student has active reviewed errors; no report	Open Recommendations → Generate	201 report appears with up to four grounded strategies	No report	One current report	No automated browser E2E test found.
E2E-UC6-02 Refresh outdated recommendations	Old report and newer corrected Sample	Open Recommendations → Refresh now	Displayed report replaced; subsequent GET is not outdated	Old report	One replaced current report	No automated browser E2E test found.
E2E-UC6-03 No active errors	All errors dismissed or no analysed errors	Select Generate/Refresh	Visible NO_ACTIVE_ERRORS failure; old report retained if present	Old/no report	Unchanged	No automated browser E2E test found.
E2E-UC6-04 External generation failure	Existing report; deterministic upstream failure	Select Refresh	Visible controlled failure; previous report remains displayed/stored	Old report	Old report unchanged	No automated browser E2E test found.

UC7 — Log In

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC7-01 Log in successfully	Known synthetic account; no session	Open login; submit valid credentials	Navigates to requested route/caseload; safe account in localStorage	No session	Client session present	No automated browser E2E test found.
E2E-UC7-02 Reject invalid login	No session	Submit unknown username; repeat with wrong password	Same inline error; remains logged out	No session	No session	No automated browser E2E test found.
E2E-UC7-03 Protected route gate	No session then valid local session	Open protected route; then log in and retry	Redirects to login when absent; allows route when present	Session absent	Session present after login	No automated browser E2E test found.

UC8 — Change Password

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC8-01 Change password successfully	Signed-in synthetic account	Open Account; enter current/strong matching new password; submit; sign out/in	Password updated shown; old login fails; new login succeeds	Old hash	New hash	No automated browser E2E test found.
E2E-UC8-02 Client rejects weak/mismatch	Signed-in account	Enter weak or mismatched new values	Submit disabled/guidance visible; no PATCH	Old hash	Old hash	No automated browser E2E test found.
E2E-UC8-03 Server rejects wrong current	Signed-in account	Enter wrong current and valid new values; submit	Inline current-password error; remains signed in; hash unchanged	Old hash	Old hash	No automated browser E2E test found.

UC9 — Reset a Forgotten Password

Test ID / Scenario	Preconditions	User Actions	Expected UI / API Result	State Before	State After	Evidence
E2E-UC9-01 Forgot and reset successfully	Known account; test email capture enabled	Request reset → open captured link → enter strong matching password	Generic sent state then Password updated; new login works	Old hash; no token	New hash; token cleared	No automated browser E2E test found.
E2E-UC9-02 Unknown username privacy	Unknown username	Request password reset	Same generic Check your email state; no message sent	No account	No account	No automated browser E2E test found.
E2E-UC9-03 Reject invalid/expired link	Invalid or expired token	Open reset link; submit strong matching password	Inline invalid/expired error; no account mutation	No valid token	No mutation	No automated browser E2E test found.
E2E-UC9-04 Reject weak/mismatch locally	Valid token	Enter weak or mismatched passwords	Submit disabled/guidance visible; token remains usable	Valid token	Valid token	No automated browser E2E test found.

5. Robustness Test Tables

Robustness cases are restricted to sequence branches, explicit validations, external boundaries and state-integrity invariants. They do not invent new product behavior.

UC1 — Upload Student Writing Sample for Analysis

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC1-01 Upload validation	No files, unsupported bytes, corrupt PDF, unknown student	Multer/toPageFiles/ createSample	400/422 safe message; all temp files cleaned	No Sample or orphan files	Teacher can correct form and retry	Partially implemented Engine unsupported/corrupt tests exist; controller upload atomicity/cleanup tests are missing.
RB-UC1-02 Multi-file atomicity	Second file fails after first validates	toPageFiles Promise.all	Whole request fails; first file is not registered	Samples remain unchanged	Retry with valid set	Not implemented No matching automated test found.
RB-UC1-03 AI output validation	Unknown category, bad confidence/location/page, missing written	validateAnalysisResult	Rejects result; retries; eventually FAILED	No invalid errors persisted	Later re-upload/retry possible	Implemented errorClassificationEngine. test.js covers all listed schema faults.
RB-UC1-04 AI timeout/retry	Hung/transient/ rate-limited Gemini call	callModelWithRetry	Bounded retries/backoff; safe FAILED result after exhaustion	Sample ends terminal, never stuck by normal failure path	Teacher sees failure and can retry	Implemented errorClassificationEngine. test.js retry/rate-limit/timeout tests.
RB-UC1-05 Polling resilience	One dropped GET then recovery; 30 non-terminal attempts	useSamplePolling	Transient failure tolerated; bounded timeout after ceiling	Server analysis continues independently	Check again restarts ceiling	Not implemented No matching automated test found.

UC2 — View Categorised Error Pattern Report

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC2-01 Sample lookup	Unknown sampleId	getSample	404; no partial report	No mutation	Return to profile	Not implemented No matching automated test found.
RB-UC2-02 Image boundary	Negative/out-of-range page index or unreadable image	getImages/ScanViewer	404 or image fallback; no raw path	No mutation	Other pages/report remain usable	Partially implemented ScanViewer image-load fallback is tested; server index/path boundary is not.
RB-UC2-03 Sensitive-path disclosure	Sample contains private imagePath	toClientSample and errors	JSON omits imagePath; image only via endpoint	Private path remains server-side	None	Partially implemented Several controller fixtures include private paths, but no direct getSample disclosure assertion found.
RB-UC2-04 Uncertainty semantics	Below-threshold score with valid category	ErrorCard/report grouping	Uncertain visual state without changing category	Sample unchanged	Teacher may confirm through UC5	Implemented ErrorCard uncertain-state test and engine threshold tests.

UC3 — View Recommended Intervention Strategies with Rationale

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC3-01 Invalid identifier	Malformed studentId	assertObjectId	400 INVALID_ID; model not queried	No mutation	Use valid student route	Implemented recommendationRoutes. test.js validates malformed IDs.
RB-UC3-02 Absent report	Valid studentId with no report	RecommendationReport. findOne	404 mapped by client to no-report state	No report	Teacher may explicitly generate	Partially implemented Client mapping/controller 404 branch has no direct automated test.
RB-UC3-03 Freshness race	Sample updated/added after generatedAt	Sample.exists freshness query	isOutdated true on read; report is not mutated	Old report retained	Teacher chooses Refresh	Implemented recommendationRoutes. test.js covers changed and newly usable evidence query.
RB-UC3-04 Read-only invariant	Current/outdated GET latest	Controller/model spies	No generateReport/engine/ upsert call	Report exists	Report unchanged	Not implemented No matching automated test found.

UC4 — Compare Error Trends Across Multiple Writing Samples

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC4-01 Date validation	Malformed, impossible, or from > to	getTrends query parser	400; no misleading trend set	No mutation	Correct dates and retry	Implemented students.integration.test.js date validation tests.
RB-UC4-02 Boundary dates	Samples late on requested to date	UTC date query	Entire to day included	No mutation	None	Implemented students.integration.test.js — includes entire requested to date.
RB-UC4-03 Count reconciliation	Dismissed and unsure errors mixed with five categories	Server aggregation + prepareTrendSamples	Dismissed excluded; unsure omitted from lines but included total	No mutation	None	Partially implemented client trends reconciliation is tested; server dismissed aggregation is not directly covered.
RB-UC4-04 Sparse selection	Zero/one sample after local filters	summariseTrends/ErrorTrendsPage	No fabricated comparison; guidance shown	No mutation	Reinclude samples/change range	Partially implemented Pure summary handles data; page guidance lacks automated test.

UC5 — Edit, Remove, Confirm, or Manually Add an Error

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC5-01 Index validation	Negative, non-integer, or out-of-range errorIndex	reclassifyError	404 Error not found; no save	Sample unchanged	Refresh report	Partially implemented sampleController.test.js covers out-of-range positive index only.
RB-UC5-02 Patch-field validation	No supported field; invalid category; non-boolean dismissed; score outside 0..1	reclassifyError	400; no partial mutation/save	Sample unchanged	Correct field and retry	Not implemented No matching automated test found.
RB-UC5-03 Manual text validation	Empty/whitespace written or invalid category	errorIndex=new branch	400; no appended error	Errors unchanged	Enter exact text/category and retry	Not implemented No matching automated test found.
RB-UC5-04 Child-text integrity	Written text includes misspelling/spacing/case	Manual append and subsequent serialization	Preserves submitted written string exactly	No manual error	Exact string stored	Partially implemented sampleController new-error test asserts a misspelling; broader whitespace/case integrity is not tested.
RB-UC5-05 Save failure	Database save rejects after in-memory patch	sample.save	Error response; UI shows failure and must not claim success	Persisted Sample unchanged	Reload authoritative Sample/retry	Not implemented No matching automated test found.

UC6 — Regenerate Intervention Recommendations

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC6-01 Missing/no-evidence input	Unknown student or no active errors	Controller/generateReport	404 or 422; no upsert	Old/no report	Add/review evidence then retry	Partially implemented generateReport no-active-errors test exists; missing-student generation test missing.
RB-UC6-02 External service failure	Azure timeout/oversize or Gemini transient/malformed output	RecommendationEngine adapters	Controlled failure; no report persistence	Old report exists	Old report remains; retry later	Partially implemented Engine faults are tested; old-report persistence invariant is not.
RB-UC6-03 Grounding integrity	Unknown evidence/worksheet IDs or unsupported category	RecommendationEngine validation	Rejects model output; no upsert	Old/no report	Fix external output/retry	Implemented recommendationEngine. test.js evidence, worksheet and category validation tests.
RB-UC6-04 Atomic replacement	Failure between generation and persistence	Controller persistence boundary	No half-superseded state; latest report stays old or becomes wholly new	One old report	Exactly one valid latest report	Partially implemented Atomic findOneAndUpdate options tested; real failure atomicity not integrated.
RB-UC6-05 First-upsert race	Concurrent POST requests on absent report	findOneAndUpdate duplicate-key branch	One winner; loser returns winner; no duplicate reports	No report	Exactly one report	Partially implemented recommendationRoutes. test.js simulates E11000 recovery.

UC7 — Log In

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC7-01 Credential validation	Missing username/password	login controller	400 validation error; no lookup	No session/account mutation	Complete fields and retry	Implemented auth.integration.test.js missing-password case.
RB-UC7-02 Enumeration resistance	Unknown username versus wrong password	login controller	Same 401 message for both	No mutation	Retry valid credentials	Implemented auth.integration.test.js asserts same message in separate cases.
RB-UC7-03 Response secrecy	Valid login account includes hash/reset fields internally	toClientAccount	Only safe account fields returned; no JWT invented	Account unchanged	Client stores safe object only	Partially implemented Successful integration test asserts exact username/email response; no broader client session payload test.
RB-UC7-04 Client-only gate limitation	Direct API call without local session	Server API boundary	Request remains public per sequence; test must not expect JWT enforcement	No session	No state change	Not implemented No matching automated test found.

UC8 — Change Password

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC8-01 Current-password secrecy	Unknown account versus wrong current password	changePassword controller	Same 401 message; no hash/save	Old hash or no account	Unchanged	Not implemented No matching automated test found.
RB-UC8-02 Server-side strength enforcement	Client bypass with weak new password	passwordPolicy in controller	400 requirements; no hash/save	Old hash	Old hash	Partially implemented Password policy is unit-tested; controller integration is missing.
RB-UC8-03 Confirmation mismatch	Strong new and different confirmation	ChangePasswordForm	Submit disabled/guidance; no API call	Old hash	Old hash	Not implemented No matching automated test found.
RB-UC8-04 Persistence failure	bcrypt succeeds but account.save rejects	account.save	Error response; old persisted hash remains	Old hash	Old hash	Not implemented No matching automated test found.
RB-UC8-05 Boundary password	Exactly 8 chars with required classes	client/server policy	Accepted consistently by both layers	Old hash	New hash after valid change	Partially implemented Server policy boundary is tested; client form consistency and controller path are not.

UC9 — Reset a Forgotten Password

Test ID / Fault Target	Fault / Input	Injection Point	Expected Handling	State Integrity	Recovery / Retry	Current Coverage
RB-UC9-01 Username privacy	Existing versus unknown username	forgotPassword	Identical generic 200; email only for existing account	No reset token	Only known account gains token	Implemented auth.integration.test.js privacy-pair tests.
RB-UC9-02 Email provider failure	Resend non-2xx/network failure	emailService after token save	Controlled server error; no false sent state	Token may already be stored per sequence order	Teacher can retry; old token expires	Not implemented No matching automated test found.
RB-UC9-03 Weak password	Valid token with weak password	resetPassword policy before lookup	400; token not consumed	Valid token	Valid token	Implemented auth.integration.test.js weak password does not consume token.
RB-UC9-04 Invalid/expired/reused token	Unknown, expired, or previously consumed token	Account.findOne token query	400 invalid or expired; no hash/save	No valid token	No state change	Partially implemented Unknown/expired is tested; explicit token reuse after success is not.
RB-UC9-05 Successful token consumption	Valid token and strong password	bcrypt + account.save	New hash saved; token and expiry cleared	Valid token; old hash	No token; new hash	Implemented auth.integration.test.js success case.

6. Sequence-to-Test Traceability and Coverage

Every planned case maps to a corrected sequence branch. Existing evidence names the closest automated test; Missing means no matching test was found. This matrix distinguishes production implementation from automated-test coverage.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
UT-UC1-01	UC1	Unit	Valid upload	Not implemented	No matching automated test found.
UT-UC1-02	UC1	Unit	Student missing or upload invalid	Not implemented	No matching automated test found.
UT-UC1-03	UC1	Unit	Polling outcomes	Not implemented	No matching automated test found.
UT-UC1-04	UC1	Unit	Valid upload	Not implemented	No matching automated test found.
UT-UC1-05	UC1	Unit	Student missing or upload invalid	Not implemented	No matching automated test found.
UT-UC1-06	UC1	Unit	Analysis succeeds	Implemented	server/services/ errorClassificationEngine. test.js — valid result, category, confidence, prompt and preservation tests.
UT-UC1-07	UC1	Unit	Gemini analysis mode	Implemented	server/services/ errorClassificationEngine. test.js — retry, malformed output, page-index, timeout and maxRetries tests.
UT-UC1-08	UC1	Unit	Analysis succeeds / fails	Implemented	server/services/ errorClassificationEngine. integration.test.js — IT-01, IT-02, IT-03 and IT-04.
E2E-UC1-01	UC1	System E2E	Upload and analyse successfully	Not implemented	No automated browser E2E test found.
E2E-UC1-02	UC1	System E2E	Reject invalid upload	Not implemented	No automated browser E2E test found.
E2E-UC1-03	UC1	System E2E	Analysis failure and polling timeout states	Not implemented	No automated browser E2E test found.
RB-UC1-01	UC1	Robustness	Upload validation	Partially implemented	Engine unsupported/corrupt tests exist; controller upload atomicity/cleanup tests are missing.
RB-UC1-02	UC1	Robustness	Multi-file atomicity	Not implemented	No matching automated test found.
RB-UC1-03	UC1	Robustness	AI output validation	Implemented	errorClassificationEngine. test.js covers all listed schema faults.
RB-UC1-04	UC1	Robustness	AI timeout/retry	Implemented	errorClassificationEngine. test.js retry/rate-limit/timeout tests.
RB-UC1-05	UC1	Robustness	Polling resilience	Not implemented	No matching automated test found.
BI-UC1-01	UC1	Bottom-up integration	Analysis succeeds	Partially implemented	errorClassificationEngine. integration.test.js IT-01/IT-04 patch model persistence; no isolated MongoDB.
BI-UC1-02	UC1	Bottom-up integration	Analysis fails	Partially implemented	errorClassificationEngine. integration.test.js IT-02 uses patched persistence; retry units are separate.
BI-UC1-03	UC1	Bottom-up integration	Valid upload	Not implemented	No matching automated test found.
BI-UC1-04	UC1	Bottom-up integration	Analysis failure	Not implemented	No matching automated test found.
BI-UC1-05	UC1	Bottom-up integration	Valid upload	Not implemented	No matching automated test found.
BI-UC1-06	UC1	Bottom-up integration	Analysis failure	Not implemented	No matching automated test found.
BI-UC1-07	UC1	Bottom-up integration	Valid upload/poll	Not implemented	No matching automated test found.
BI-UC1-08	UC1	Bottom-up integration	Upload invalid	Not implemented	No matching automated test found.
BI-UC1-09	UC1	Bottom-up integration	Upload and poll success	Not implemented	No matching automated test found.
BI-UC1-10	UC1	Bottom-up integration	Analysis failure	Partially implemented	client/test/status.test. js covers terminal labels only; page/hook/full stack are untested.
UT-UC2-01	UC2	Unit	Sample exists	Implemented	client/test/ SampleReportPage.test.js — grouping, dismissed-error and filter tests.
UT-UC2-02	UC2	Unit	confidenceScore < confidenceThreshold	Implemented	client/test/ErrorCard. test.js — uncertain state flags below-threshold confidence and confirms via onConfirm.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
UT-UC2-03	UC2	Unit	For each displayed scan page	Implemented	client/test/ScanViewer. test.js — box filtering, click, thumbnails, zoom, overlay and image-failure tests.
UT-UC2-04	UC2	Unit	Sample exists	Not implemented	No matching automated test found.
UT-UC2-05	UC2	Unit	Sample not found	Not implemented	No matching automated test found.
UT-UC2-06	UC2	Unit	Display scan page	Not implemented	No matching automated test found.
UT-UC2-07	UC2	Unit	compute Statistics(non-dismissed errors)	Not implemented	No matching automated test found.
E2E-UC2-01	UC2	System E2E	View categorised report	Not implemented	No automated browser E2E test found.
E2E-UC2-02	UC2	System E2E	Missing report or image	Not implemented	No automated browser E2E test found.
RB-UC2-01	UC2	Robustness	Sample lookup	Not implemented	No matching automated test found.
RB-UC2-02	UC2	Robustness	Image boundary	Partially implemented	ScanViewer image-load fallback is tested; server index/path boundary is not.
RB-UC2-03	UC2	Robustness	Sensitive-path disclosure	Partially implemented	Several controller fixtures include private paths, but no direct getSample disclosure assertion found.
RB-UC2-04	UC2	Robustness	Uncertainty semantics	Implemented	ErrorCard uncertain-state test and engine threshold tests.
BI-UC2-01	UC2	Bottom-up integration	Sample exists	Not implemented	No matching automated test found.
BI-UC2-02	UC2	Bottom-up integration	Display scan page	Not implemented	No matching automated test found.
BI-UC2-03	UC2	Bottom-up integration	Sample exists	Not implemented	No matching automated test found.
BI-UC2-04	UC2	Bottom-up integration	Sample/image missing	Not implemented	No matching automated test found.
BI-UC2-05	UC2	Bottom-up integration	View categorised report	Not implemented	No matching automated test found.
BI-UC2-06	UC2	Bottom-up integration	Sample/image missing	Not implemented	No matching automated test found.
BI-UC2-07	UC2	Bottom-up integration	Sample exists	Not implemented	No matching automated test found.
BI-UC2-08	UC2	Bottom-up integration	Sample missing	Not implemented	No matching automated test found.
BI-UC2-09	UC2	Bottom-up integration	Sample exists	Partially implemented	SampleReportPage/ ScanViewer/ErrorCard tests use mocked API responses, not integrated core.
BI-UC2-10	UC2	Bottom-up integration	Sample missing	Not implemented	No matching automated test found.
UT-UC3-01	UC3	Unit	Current report exists	Not implemented	No matching automated test found.
UT-UC3-02	UC3	Unit	No recommendation report exists	Not implemented	No matching automated test found.
UT-UC3-03	UC3	Unit	isOutdated = true	Not implemented	No matching automated test found.
UT-UC3-04	UC3	Unit	Current report exists	Partially implemented	server/test/ recommendationRoutes. test.js tests stale=true but not current=false or explicit no-write behavior.
UT-UC3-05	UC3	Unit	isOutdated = true	Implemented	server/test/ recommendationRoutes. test.js — latest report is marked outdated when usable sample evidence changed.
UT-UC3-06	UC3	Unit	No recommendation report exists	Partially implemented	server/test/ recommendationRoutes. test.js covers malformed ID; no exact latest-report 404 test found.
E2E-UC3-01	UC3	System E2E	View current recommendations	Not implemented	No automated browser E2E test found.
E2E-UC3-02	UC3	System E2E	View no-report state	Not implemented	No automated browser E2E test found.
E2E-UC3-03	UC3	System E2E	View outdated report	Not implemented	No automated browser E2E test found.
RB-UC3-01	UC3	Robustness	Invalid identifier	Implemented	recommendationRoutes. test.js validates malformed IDs.
RB-UC3-02	UC3	Robustness	Absent report	Partially implemented	Client mapping/controller 404 branch has no direct automated test.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
RB-UC3-03	UC3	Robustness	Freshness race	Implemented	recommendationRoutes. test.js covers changed and newly usable evidence query.
RB-UC3-04	UC3	Robustness	Read-only invariant	Not implemented	No matching automated test found.
BI-UC3-01	UC3	Bottom-up integration	Current report exists	Partially implemented	recommendationReport. test.js checks schema shape only; no isolated MongoDB.
BI-UC3-02	UC3	Bottom-up integration	Evidence changed	Partially implemented	recommendationRoutes. test.js covers freshness with mocked models.
BI-UC3-03	UC3	Bottom-up integration	Current report exists	Partially implemented	recommendationRoutes. test.js covers stale=true only with mocked models.
BI-UC3-04	UC3	Bottom-up integration	No report exists	Not implemented	No matching automated test found.
BI-UC3-05	UC3	Bottom-up integration	Evidence changed	Partially implemented	recommendationRoutes. test.js directly covers changed/new evidence using stubs.
BI-UC3-06	UC3	Bottom-up integration	Current report exists	Partially implemented	Current route tests invoke controller with mocked models; no real route/database stack.
BI-UC3-07	UC3	Bottom-up integration	No report exists	Not implemented	No matching automated test found.
BI-UC3-08	UC3	Bottom-up integration	Current report exists	Not implemented	No matching automated test found.
BI-UC3-09	UC3	Bottom-up integration	No report exists	Not implemented	No matching automated test found.
BI-UC3-10	UC3	Bottom-up integration	Current report exists	Not implemented	No matching automated test found.
BI-UC3-11	UC3	Bottom-up integration	No report exists	Not implemented	No matching automated test found.
BI-UC3-12	UC3	Bottom-up integration	Evidence changed	Not implemented	No matching automated test found.
UT-UC4-01	UC4	Unit	Prepare trend samples	Implemented	client/test/trends.test. js — chronological order and five-category reconciliation tests.
UT-UC4-02	UC4	Unit	Choose date range and subset	Partially implemented	client/test/trends.test. js covers preset range and explicit exclusion; custom-range helper has no direct test.
UT-UC4-03	UC4	Unit	At least 2 included samples	Implemented	client/test/trends.test. js — tie order and all three comparison states.
UT-UC4-04	UC4	Unit	Fewer than 2 included samples	Not implemented	No matching automated test found.
UT-UC4-05	UC4	Unit	Student exists	Partially implemented	server/test/students. integration.test.js covers inclusive to date; no full aggregation/dismissed test found.
UT-UC4-06	UC4	Unit	Validate optional from/to	Implemented	server/test/students. integration.test.js — malformed, impossible and reversed date tests.
UT-UC4-07	UC4	Unit	Student missing	Implemented	server/test/students. integration.test.js — GET trends reports an unknown student as 404.
E2E-UC4-01	UC4	System E2E	Compare multiple samples	Not implemented	No automated browser E2E test found.
E2E-UC4-02	UC4	System E2E	Insufficient comparison	Not implemented	No automated browser E2E test found.
E2E-UC4-03	UC4	System E2E	Missing student	Not implemented	No automated browser E2E test found.
RB-UC4-01	UC4	Robustness	Date validation	Implemented	students.integration. test.js date validation tests.
RB-UC4-02	UC4	Robustness	Boundary dates	Implemented	students.integration. test.js — includes entire requested to date.
RB-UC4-03	UC4	Robustness	Count reconciliation	Partially implemented	client trends reconciliation is tested; server dismissed aggregation is not directly covered.
RB-UC4-04	UC4	Robustness	Sparse selection	Partially implemented	Pure summary handles data; page guidance lacks automated test.
BI-UC4-01	UC4	Bottom-up integration	Student exists	Partially implemented	students.integration. test.js asserts status/date query with stubbed models, not isolated MongoDB.
BI-UC4-02	UC4	Bottom-up integration	Student missing	Partially implemented	students.integration. test.js covers 404 path with model stubs.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
BI-UC4-03	UC4	Bottom-up integration	Trend aggregation	Partially implemented	Controller tests do not use non-empty live+dismissed persistence; related helpers are separate.
BI-UC4-04	UC4	Bottom-up integration	Invalid optional dates	Partially implemented	students.integration. test.js covers malformed/impossible/ reversed dates with stubs.
BI-UC4-05	UC4	Bottom-up integration	Student exists	Partially implemented	students.integration. test.js exercises HTTP/controller boundary with fake model statics.
BI-UC4-06	UC4	Bottom-up integration	Student/date invalid	Partially implemented	Existing route tests use stubbed persistence.
BI-UC4-07	UC4	Bottom-up integration	Student exists	Not implemented	No matching automated test found.
BI-UC4-08	UC4	Bottom-up integration	Student missing	Not implemented	No matching automated test found.
BI-UC4-09	UC4	Bottom-up integration	At least 2 included samples	Partially implemented	client/test/trends.test. js covers pure helpers only; page/full stack is untested.
BI-UC4-10	UC4	Bottom-up integration	Fewer than 2 samples	Not implemented	No matching automated test found.
UT-UC5-01	UC5	Unit	Reclassify existing error	Implemented	client/test/ErrorCard. test.js — reclassify panel saves a different category.
UT-UC5-02	UC5	Unit	Remove or restore existing error	Implemented	client/test/ErrorCard. test.js — remove confirmation and removed-state restore tests.
UT-UC5-03	UC5	Unit	Confirm uncertain error	Implemented	client/test/ErrorCard. test.js — uncertain state confirmation.
UT-UC5-04	UC5	Unit	Record an error missed by AI	Implemented	client/test/ SampleReportPage.test.js — educator can submit an error the AI missed.
UT-UC5-05	UC5	Unit	Valid existing-error patch	Not implemented	No matching automated test found.
UT-UC5-06	UC5	Unit	Remove existing error	Implemented	server/test/ sampleController.test.js — dismissal-only error updates save once and return full sample.
UT-UC5-07	UC5	Unit	Confirm uncertain error	Implemented	server/test/ sampleController.test.js — confidence-only update preserves other fields.
UT-UC5-08	UC5	Unit	Valid new manual error	Implemented	server/test/ sampleController.test.js — error index new appends educator-identified error.
UT-UC5-09	UC5	Unit	Invalid patch	Partially implemented	server/test/ sampleController.test.js covers out-of-range index only; other validation branches are missing.
E2E-UC5-01	UC5	System E2E	Correct existing AI tags	Not implemented	No automated browser E2E test found.
E2E-UC5-02	UC5	System E2E	Restore removed tag	Not implemented	No automated browser E2E test found.
E2E-UC5-03	UC5	System E2E	Add an AI-missed error	Not implemented	No automated browser E2E test found.
E2E-UC5-04	UC5	System E2E	Rejected correction	Not implemented	No automated browser E2E test found.
RB-UC5-01	UC5	Robustness	Index validation	Partially implemented	sampleController.test.js covers out-of-range positive index only.
RB-UC5-02	UC5	Robustness	Patch-field validation	Not implemented	No matching automated test found.
RB-UC5-03	UC5	Robustness	Manual text validation	Not implemented	No matching automated test found.
RB-UC5-04	UC5	Robustness	Child-text integrity	Partially implemented	sampleController new-error test asserts a misspelling; broader whitespace/case integrity is not tested.
RB-UC5-05	UC5	Robustness	Save failure	Not implemented	No matching automated test found.
BI-UC5-01	UC5	Bottom-up integration	Valid existing patch	Partially implemented	sampleController.test.js uses a mocked Sample document; no isolated MongoDB.
BI-UC5-02	UC5	Bottom-up integration	Valid manual error	Partially implemented	sampleController.test.js covers new index with mocked persistence.
BI-UC5-03	UC5	Bottom-up integration	Valid patch alternatives	Partially implemented	sampleController.test.js covers dismiss, confidence and new; reclassify/restore remain incomplete.
BI-UC5-04	UC5	Bottom-up integration	Invalid patch	Partially implemented	Only out-of-range index has direct controller coverage.
BI-UC5-05	UC5	Bottom-up integration	Valid patch alternatives	Partially implemented	Controller tests do not prove the real route plus isolated database boundary.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
BI-UC5-06	UC5	Bottom-up integration	Invalid patch	Not implemented	No matching automated test found.
BI-UC5-07	UC5	Bottom-up integration	Valid patch alternatives	Not implemented	No matching automated test found.
BI-UC5-08	UC5	Bottom-up integration	Invalid/missing	Not implemented	No matching automated test found.
BI-UC5-09	UC5	Bottom-up integration	Valid correction actions	Partially implemented	ErrorCard/ SampleReportPage tests mock updateSampleError; not full stack.
BI-UC5-10	UC5	Bottom-up integration	Save failure	Not implemented	No matching automated test found.
UT-UC6-01	UC6	Unit	Generate or Refresh recommendations	Not implemented	No matching automated test found.
UT-UC6-02	UC6	Unit	Active evidence exists	Implemented	server/test/ generateReport.test.js — uses all analysed samples, excludes private identity/path data.
UT-UC6-03	UC6	Unit	No active errors	Implemented	server/test/ generateReport.test.js — rejects students without analysed errors.
UT-UC6-04	UC6	Unit	Gemini recommendation mode	Implemented	server/test/ recommendationEngine. test.js — grounding, mapping, validation and mock-selection tests.
UT-UC6-05	UC6	Unit	Generation failure	Implemented	server/test/ recommendationEngine. test.js — retry, malformed JSON, timeout/size controls.
UT-UC6-06	UC6	Unit	Upsert succeeds	Implemented	server/test/ recommendationRoutes. test.js — atomically upserts latest-only report.
UT-UC6-07	UC6	Unit	Student missing	Not implemented	No matching automated test found.
UT-UC6-08	UC6	Unit	Duplicate-key race	Implemented	server/test/ recommendationRoutes. test.js — recovers when concurrent first upsert wins.
UT-UC6-09	UC6	Unit	Generation failure	Not implemented	No matching automated test found.
E2E-UC6-01	UC6	System E2E	Generate first recommendations	Not implemented	No automated browser E2E test found.
E2E-UC6-02	UC6	System E2E	Refresh outdated recommendations	Not implemented	No automated browser E2E test found.
E2E-UC6-03	UC6	System E2E	No active errors	Not implemented	No automated browser E2E test found.
E2E-UC6-04	UC6	System E2E	External generation failure	Not implemented	No automated browser E2E test found.
RB-UC6-01	UC6	Robustness	Missing/no-evidence input	Partially implemented	generateReport no-active-errors test exists; missing-student generation test missing.
RB-UC6-02	UC6	Robustness	External service failure	Partially implemented	Engine faults are tested; old-report persistence invariant is not.
RB-UC6-03	UC6	Robustness	Grounding integrity	Implemented	recommendationEngine. test.js evidence, worksheet and category validation tests.
RB-UC6-04	UC6	Robustness	Atomic replacement	Partially implemented	Atomic findOneAndUpdate options tested; real failure atomicity not integrated.
RB-UC6-05	UC6	Robustness	First-upsert race	Partially implemented	recommendationRoutes. test.js simulates E11000 recovery.
BI-UC6-01	UC6	Bottom-up integration	Active evidence exists	Partially implemented	generateReport/ recommendationEngine tests use mocked models/external calls; no isolated MongoDB.
BI-UC6-02	UC6	Bottom-up integration	No active errors	Partially implemented	generateReport.test.js directly covers no-error branch without real persistence.
BI-UC6-03	UC6	Bottom-up integration	Generation succeeds	Partially implemented	recommendationRoutes. test.js covers atomic upsert with mocked models.
BI-UC6-04	UC6	Bottom-up integration	Generation fails	Not implemented	No matching automated test found.
BI-UC6-05	UC6	Bottom-up integration	Concurrent first upsert	Partially implemented	recommendationRoutes. test.js covers E11000 recovery using stubs.
BI-UC6-06	UC6	Bottom-up integration	Generation succeeds	Partially implemented	Existing tests cover controller/app seam with mocked persistence, not isolated MongoDB.
BI-UC6-07	UC6	Bottom-up integration	Generation fails	Partially implemented	No-active branch covered; old-report retention is not directly asserted.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
BI-UC6-08	UC6	Bottom-up integration	Generation succeeds	Not implemented	No matching automated test found.
BI-UC6-09	UC6	Bottom-up integration	Generation fails	Not implemented	No matching automated test found.
BI-UC6-10	UC6	Bottom-up integration	Generation succeeds	Not implemented	No matching automated test found.
BI-UC6-11	UC6	Bottom-up integration	Generation fails	Not implemented	No matching automated test found.
UT-UC7-01	UC7	Unit	Credentials valid	Not implemented	No matching automated test found.
UT-UC7-02	UC7	Unit	Credentials invalid	Not implemented	No matching automated test found.
UT-UC7-03	UC7	Unit	Credentials valid	Implemented	server/test/auth. integration.test.js — successful login returns username and email.
UT-UC7-04	UC7	Unit	Username or password missing	Implemented	server/test/auth. integration.test.js — missing password through error middleware.
UT-UC7-05	UC7	Unit	Account missing or password mismatch	Implemented	server/test/auth. integration.test.js — unknown username and wrong password tests.
UT-UC7-06	UC7	Unit	Client-side route gate	Partially implemented	client/test/RequireAuth. test.js covers redirect/allow; saveSession itself and LoginPage handoff are not tested.
E2E-UC7-01	UC7	System E2E	Log in successfully	Not implemented	No automated browser E2E test found.
E2E-UC7-02	UC7	System E2E	Reject invalid login	Not implemented	No automated browser E2E test found.
E2E-UC7-03	UC7	System E2E	Protected route gate	Not implemented	No automated browser E2E test found.
RB-UC7-01	UC7	Robustness	Credential validation	Implemented	auth.integration.test.js missing-password case.
RB-UC7-02	UC7	Robustness	Enumeration resistance	Implemented	auth.integration.test.js asserts same message in separate cases.
RB-UC7-03	UC7	Robustness	Response secrecy	Partially implemented	Successful integration test asserts exact username/email response; no broader client session payload test.
RB-UC7-04	UC7	Robustness	Client-only gate limitation	Not implemented	No matching automated test found.
BI-UC7-01	UC7	Bottom-up integration	Credentials valid	Partially implemented	auth.integration.test.js uses patched Account methods; bcrypt is real but MongoDB is not.
BI-UC7-02	UC7	Bottom-up integration	Credentials invalid	Partially implemented	auth.integration.test.js covers unknown/wrong cases with patched model methods.
BI-UC7-03	UC7	Bottom-up integration	Credentials valid	Partially implemented	auth.integration.test.js directly asserts safe object with patched persistence.
BI-UC7-04	UC7	Bottom-up integration	Credentials invalid	Partially implemented	auth.integration.test.js covers both generic 401 branches.
BI-UC7-05	UC7	Bottom-up integration	Credentials valid	Partially implemented	auth.integration.test.js exercises Express boundary but not real MongoDB.
BI-UC7-06	UC7	Bottom-up integration	Credentials invalid	Partially implemented	Missing-password/unknown/ wrong tests exist; other missing-field combinations do not.
BI-UC7-07	UC7	Bottom-up integration	Credentials valid	Not implemented	No matching automated test found.
BI-UC7-08	UC7	Bottom-up integration	Credentials invalid	Not implemented	No matching automated test found.
BI-UC7-09	UC7	Bottom-up integration	Credentials valid	Not implemented	No matching automated test found.
BI-UC7-10	UC7	Bottom-up integration	Credentials invalid	Partially implemented	RequireAuth tests route gating only; LoginPage/session full integration is missing.
UT-UC8-01	UC8	Unit	Client validation passes	Not implemented	No matching automated test found.
UT-UC8-02	UC8	Unit	Client validation fails	Not implemented	No matching automated test found.
UT-UC8-03	UC8	Unit	Server password policy	Implemented	server/test/ passwordPolicy.test.js — valid, every unmet rule, each single-rule failure and boundary tests.
UT-UC8-04	UC8	Unit	New password valid	Not implemented	No matching automated test found.
UT-UC8-05	UC8	Unit	Current password incorrect	Not implemented	No matching automated test found.
UT-UC8-06	UC8	Unit	Server password policy fails	Not implemented	No matching automated test found.
E2E-UC8-01	UC8	System E2E	Change password successfully	Not implemented	No automated browser E2E test found.
E2E-UC8-02	UC8	System E2E	Client rejects weak/mismatch	Not implemented	No automated browser E2E test found.
E2E-UC8-03	UC8	System E2E	Server rejects wrong current	Not implemented	No automated browser E2E test found.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
RB-UC8-01	UC8	Robustness	Current-password secrecy	Not implemented	No matching automated test found.
RB-UC8-02	UC8	Robustness	Server-side strength enforcement	Partially implemented	Password policy is unit-tested; controller integration is missing.
RB-UC8-03	UC8	Robustness	Confirmation mismatch	Not implemented	No matching automated test found.
RB-UC8-04	UC8	Robustness	Persistence failure	Not implemented	No matching automated test found.
RB-UC8-05	UC8	Robustness	Boundary password	Partially implemented	Server policy boundary is tested; client form consistency and controller path are not.
BI-UC8-01	UC8	Bottom-up integration	Passwords valid	Not implemented	No matching automated test found.
BI-UC8-02	UC8	Bottom-up integration	Server password policy fails	Partially implemented	passwordPolicy.test.js covers rules/boundaries only; no real account persistence.
BI-UC8-03	UC8	Bottom-up integration	Passwords valid	Not implemented	No matching automated test found.
BI-UC8-04	UC8	Bottom-up integration	Current password invalid	Not implemented	No matching automated test found.
BI-UC8-05	UC8	Bottom-up integration	Server policy fails	Partially implemented	Pure password policy tests exist; controller invocation is untested.
BI-UC8-06	UC8	Bottom-up integration	Passwords valid	Not implemented	No matching automated test found.
BI-UC8-07	UC8	Bottom-up integration	Password invalid	Not implemented	No matching automated test found.
BI-UC8-08	UC8	Bottom-up integration	Passwords valid	Not implemented	No matching automated test found.
BI-UC8-09	UC8	Bottom-up integration	Password invalid	Not implemented	No matching automated test found.
BI-UC8-10	UC8	Bottom-up integration	Passwords valid	Not implemented	No matching automated test found.
BI-UC8-11	UC8	Bottom-up integration	Client/server validation fails	Not implemented	No matching automated test found.
UT-UC9-01	UC9	Unit	Reset request accepted	Not implemented	No matching automated test found.
UT-UC9-02	UC9	Unit	Email error	Not implemented	No matching automated test found.
UT-UC9-03	UC9	Unit	Account exists	Implemented	server/test/auth. integration.test.js — sends reset email and stores token.
UT-UC9-04	UC9	Unit	Account missing	Implemented	server/test/auth. integration.test.js — identical response for unknown username and no email.
UT-UC9-05	UC9	Unit	Client validation fails	Not implemented	No matching automated test found.
UT-UC9-06	UC9	Unit	Token valid	Implemented	server/test/auth. integration.test.js — updates password and clears token on success.
UT-UC9-07	UC9	Unit	Server password policy fails	Implemented	server/test/auth. integration.test.js — weak password does not consume token.
UT-UC9-08	UC9	Unit	Token invalid or expired	Partially implemented	auth.integration.test.js test name mentions an expired token, but the fixture directly exercises only an unknown token.
UT-UC9-09	UC9	Unit	Resend accepted or error	Not implemented	No matching automated test found.
E2E-UC9-01	UC9	System E2E	Forgot and reset successfully	Not implemented	No automated browser E2E test found.
E2E-UC9-02	UC9	System E2E	Unknown username privacy	Not implemented	No automated browser E2E test found.
E2E-UC9-03	UC9	System E2E	Reject invalid/expired link	Not implemented	No automated browser E2E test found.
E2E-UC9-04	UC9	System E2E	Reject weak/mismatch locally	Not implemented	No automated browser E2E test found.
RB-UC9-01	UC9	Robustness	Username privacy	Implemented	auth.integration.test.js privacy-pair tests.
RB-UC9-02	UC9	Robustness	Email provider failure	Not implemented	No matching automated test found.
RB-UC9-03	UC9	Robustness	Weak password	Implemented	auth.integration.test.js weak password does not consume token.
RB-UC9-04	UC9	Robustness	Invalid/expired/reused token	Partially implemented	Unknown/expired is tested; explicit token reuse after success is not.
RB-UC9-05	UC9	Robustness	Successful token consumption	Implemented	auth.integration.test.js success case.
BI-UC9-01	UC9	Bottom-up integration	Known account reset request	Partially implemented	auth.integration.test.js patches Account and mocks email service; no isolated MongoDB/Resend adapter.
BI-UC9-02	UC9	Bottom-up integration	Unknown account reset request	Partially implemented	auth.integration.test.js covers identical unknown response through controller with stubs.

Test ID	Use Case	Level	Sequence Branch	Coverage Status	Existing Automated-Test Evidence
BI-UC9-03	UC9	Bottom-up integration	Token valid	Partially implemented	auth.integration.test.js covers success with patched Account methods.
BI-UC9-04	UC9	Bottom-up integration	Anti-enumeration branch	Partially implemented	auth.integration.test.js directly covers both branches with mocks.
BI-UC9-05	UC9	Bottom-up integration	Token valid	Partially implemented	auth.integration.test.js covers valid reset with stubbed persistence.
BI-UC9-06	UC9	Bottom-up integration	Token/policy invalid	Partially implemented	Weak/unknown token covered; named expired-token test fixture does not exercise expiry.
BI-UC9-07	UC9	Bottom-up integration	Anti-enumeration branch	Partially implemented	auth.integration.test.js exercises Express route with patched persistence and email mock.
BI-UC9-08	UC9	Bottom-up integration	Token valid	Partially implemented	Existing auth integration test uses patched Account methods.
BI-UC9-09	UC9	Bottom-up integration	Token/policy invalid	Partially implemented	Weak and unknown token paths covered; real expiry boundary is missing.
BI-UC9-10	UC9	Bottom-up integration	Anti-enumeration branch	Not implemented	No matching automated test found.
BI-UC9-11	UC9	Bottom-up integration	Token valid	Not implemented	No matching automated test found.
BI-UC9-12	UC9	Bottom-up integration	Token/policy invalid	Not implemented	No matching automated test found.
BI-UC9-13	UC9	Bottom-up integration	Anti-enumeration branch	Not implemented	No matching automated test found.
BI-UC9-14	UC9	Bottom-up integration	Token valid	Not implemented	No matching automated test found.
BI-UC9-15	UC9	Bottom-up integration	Token/policy invalid	Not implemented	No matching automated test found.

Coverage summary by use case and level

Use Case	Level	Implemented	Partial	Missing	Conflict	Total
UC1	Unit	3	0	5	0	8
UC1	Bottom-up integration	0	3	7	0	10
UC1	System E2E	0	0	3	0	3
UC1	Robustness	2	1	2	0	5
UC2	Unit	3	0	4	0	7
UC2	Bottom-up integration	0	1	9	0	10
UC2	System E2E	0	0	2	0	2
UC2	Robustness	1	2	1	0	4
UC3	Unit	1	2	3	0	6
UC3	Bottom-up integration	0	5	7	0	12
UC3	System E2E	0	0	3	0	3
UC3	Robustness	2	1	1	0	4
UC4	Unit	4	2	1	0	7
UC4	Bottom-up integration	0	7	3	0	10
UC4	System E2E	0	0	3	0	3
UC4	Robustness	2	2	0	0	4
UC5	Unit	7	1	1	0	9
UC5	Bottom-up integration	0	6	4	0	10
UC5	System E2E	0	0	4	0	4
UC5	Robustness	0	2	3	0	5
UC6	Unit	6	0	3	0	9
UC6	Bottom-up integration	0	6	5	0	11
UC6	System E2E	0	0	4	0	4
UC6	Robustness	1	4	0	0	5
UC7	Unit	3	1	2	0	6
UC7	Bottom-up integration	0	7	3	0	10
UC7	System E2E	0	0	3	0	3
UC7	Robustness	2	1	1	0	4
UC8	Unit	1	0	5	0	6
UC8	Bottom-up integration	0	2	9	0	11
UC8	System E2E	0	0	3	0	3
UC8	Robustness	0	2	3	0	5
UC9	Unit	4	1	4	0	9
UC9	Bottom-up integration	0	9	6	0	15
UC9	System E2E	0	0	4	0	4
UC9	Robustness	3	1	1	0	5

Key coverage conclusions

- Strongest current coverage: AI error-result validation, recommendation-engine safety checks, trend pure functions, login/reset server branches, and UC5 embedded-error updates.
- Largest gaps: automated browser E2E, UploadSamplePage/useSamplePolling, report-read controllers, recommendations UI, AccountPage change-password flow, and Forgot/Reset client pages.
- Current files named integration often use model stubs. They are therefore Partial against this plan's isolated-database integration boundary.
- No current automated test was found that conflicts with the corrected sequence diagrams.
- UC3 must remain read-only; UC6 must generate before one atomic latest-only upsert; UC5 uncertainty remains confidence state rather than a sixth educator category.